Collective Intent Drift:

Weakest-Link Cascades and a Disorder-Induced Critical Endpoint in Populations of Coupled Agents

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Abstract

A companion diagnosis shows that a single agent optimised against a monitor undergoes a saddle–node bifurcation—an abrupt, irreversible flip from loyal behaviour to reward hacking—once the monitor's steepness exceeds a threshold. Multi-agent systems (debate, self-play, agent swarms) couple many such agents through shared rewards or mutual monitoring. We ask whether single-agent bistability aggregates into a *collective* avalanche, what sets its threshold, and how it depends on the heterogeneity of the population. We show three things. *(i)* The population undergoes a mean-field first-order transition—a sharp, hysteretic collective flip at a critical coupling $\kappa^\star$—mean-field sharp, with finite-N realisations rounded at width $O(N^{-1/2})$. *(ii)* $\kappa^\star$ is governed by the *lower tail* of the vulnerability distribution, not the mean: an exact self-consistency tangency whose leading behaviour is a bootstrap logarithm $\kappa^\star \approx -\lambda^{-1} \ln f$ in the rotten-apple fraction f. *(iii)* The order of the collective transition is itself controlled by the population's *heterogeneity*: as the spread of vulnerability grows, the first-order jump shrinks and vanishes at a critical spread CV_c, where the two mean-field folds merge in a cusp—the same cusp singularity that governs the single-agent fold—beyond which the transition is continuous. By the cusp normal form the jump vanishes as $(\mathrm{CV}_c - \mathrm{CV})^{1/2}$. The cascade phenomena are classical (mean-field first-order transitions; Granovetter/Watts threshold cascades; the disorder-rounding of first-order transitions); the contribution is the microfoundation (each agent's threshold is the single-agent fold), the mapping to multi-agent reward hacking, and the actionable, counter-intuitive law: a *homogeneous* fleet flips as one cliff, while heterogeneity softens the collective catastrophe into a crossover. Harden the weakest agents, not the average—but expect a diverse fleet to fail gradually rather than all at once.

1 Introduction

Deployed AI increasingly runs as *populations* of interacting agents—debating, grading one another, sharing tools and rewards. Empirically such populations exhibit collective misbehaviour: agents in a shared economy develop deception without any reward for it, and misalignment can spread between models (Fraga-Gonçalves et al., 2025; Betley et al., 2026). Are these collective failures a smooth aggregate of independent lapses, or a genuine phase transition—and if so, of what order?

We build on a companion diagnosis in which a single agent's evasion coordinate $\rho \in [0, 1]$ (its hack footprint hidden from a monitor) obeys a best-response flow $\dot{\rho} = \mu\, \partial_\rho \Pi(\rho; w)$ with a saddle–node fold: past a critical monitor steepness the loyal→hacking flip is abrupt and hysteretic, at loyal-fold stakes w_+. Here we couple N such agents through a shared incentive—an agent's effective stakes rise with the mean hacking of its neighbourhood (herd pressure, or a monitor diluted across a hacking crowd)—and ask when, and how sharply, the population avalanches.

Contributions.

1. **Mean-field avalanche** (Theorem 1): the homogeneous order parameter obeys a self-consistency whose loyal branch vanishes at a critical coupling $\kappa^\star$; the collective flip is a sharp, hysteretic first-order transition (mean-field sharp; finite-N rounded at $O(N^{-1/2})$).

2. **Weakest-link law** (Theorem 2): the exact threshold is a self-consistency tangency governed by the majority fold; its leading small-f behaviour is a bootstrap logarithm $\kappa^\star \approx -\lambda^{-1} \ln f + \kappa_0$, tail-dominated. Under k-regular coupling $\kappa^\star(k) = a \ln k + b$ grows with degree (Corollary 1)—sparse coupling is the most dangerous.

3. **Disorder-induced critical endpoint** (Theorem 3): the *order* of the collective transition is set by the population's heterogeneity. As the vulnerability spread CV grows, the first-order jump $\Delta m(\mathrm{CV})$ shrinks and vanishes at a critical spread CV_c, where the two mean-field folds merge in a cusp; by the cusp normal form $\Delta m \sim (\mathrm{CV}_c - \mathrm{CV})^{1/2}$. Beyond CV_c the transition is continuous.

Positioning. None of the cascade mathematics is new. Mean-field first-order transitions, Granovetter's threshold model, Watts's cascade window, and bootstrap percolation are classical (Granovetter, 1978; Watts, 2002); the softening of a first-order transition by heterogeneity is *analogous* to the disorder-rounding of first-order transitions studied in statistical physics (Imry and Ma, 1975; Aizenman and Wehr, 1989) (though our mechanism is a mean-field spread of fold locations, not a random-field argument, so we draw the connection as an analogy rather than a direct instance). Our contribution is to give these a *microfoundation*—each agent's threshold is the single-agent saddle–node of the companion, so the macroscopic cascade inherits the microscopic bifurcation, and the endpoint cusp is the same singularity as the single-agent fold—and to map the result onto multi-agent reward hacking, yielding actionable laws. We work with a surrogate; §9 states its scope.

2 Related work

Multi-agent misalignment. Populations of agents show emergent deception and cross-model contagion, so far documented empirically (Fraga-Gonçalves et al., 2025; Betley et al., 2026), and agents can collude covertly (Motwani et al., 2024); taxonomies of multi-agent risk are emerging (Hammond et al., 2025). We add a mean-field phase-transition model with a microfounded threshold, a tail law, and a transition-order phase diagram.

Threshold and cascade models; disorder rounding. Granovetter's threshold model (Granovetter, 1978), Watts's cascade model (Watts, 2002), informational cascades (Bikhchandani et al., 1992), and generalized contagion (Dodds and Watts, 2004) explain how a small seed triggers population-wide adoption within a connectivity window; our weakest-link logarithm and degree law are these phenomena with a bifurcation microfoundation. That quenched disorder can round a first-order transition into a continuous one is a classical theme in statistical physics (Imry and Ma, 1975; Aizenman and Wehr, 1989); our Theorem 3 is *analogous*—the heterogeneity of vulnerability spreads the individual folds and merges them in a cusp—rather than a direct instance of the random-field mechanism.

3 Model and microfoundation

Population. N agents with evasion coordinates $\rho_i \in [0,1]$; agent i has cost scale $c_i > 0$ (hiding cost $c_i C_0$ for a fixed convex shape C_0), drawn i.i.d. from a law ν. Coupling raises each agent's effective stakes with the mean hacking of its neighbourhood N_i:

$$\dot{\rho}_i = \mu\, \partial_{\rho_i} \Pi\big(\rho_i;\, w_0 + \kappa\, \bar{\rho}_i\big), \qquad \bar{\rho}_i = \tfrac{1}{|N_i|} \textstyle\sum_{j \in N_i} \rho_j, \tag{1}$$

where $\Pi(\rho; w) = w(1 - A(\rho)) - c_i C_0(\rho)$ is the single-agent payoff, w_0 base stakes, κ the coupling. Write $\rho_{\text{eq}}(W; c)$ for the single-agent stable equilibrium reached by *loyal-branch continuation*: starting from the loyal state at small W and following the stable branch as W increases, so ρ_{eq} is the lower stable root of $W h(\rho) = c C_0'(\rho)$ until the loyal fold $w_+(c)$, and the hacking root thereafter. This upward-continuation convention (matching a slowly ramped task) fixes the branch unambiguously in the bistable window; $w_+(c)$ is the loyal-fold stakes at which the loyal root disappears.

Lemma 1 (Fold linear in cost scale)**.** *$w_+(c) = c\, w_+(1)$ for any fixed convex cost shape C_0. Hence the vulnerability axis is one-dimensional, and an agent* self-ignites *(hacks at $\kappa = 0$) iff $w_0 > w_+(c)$, i.e. $c < c_* := w_0 / w_+(1)$.*

Proof. $w_+(c) = \max_{\rho < \rho^\star} \{c\, C_0'(\rho)/h(\rho)\} = c \max_{\rho < \rho^\star} \{C_0'(\rho)/h(\rho)\} = c\, w_+(1)$ with $h = -A'$; self-ignition is $w_0 > w_+(c)$. □

4 The collective avalanche

Theorem 1 (Mean-field avalanche is a first-order transition)**.** *In the all-to-all limit the homogeneous order parameter $m = \langle \rho \rangle$ obeys the self-consistency $m = \Phi(m) := \mathbb{E}_{c \sim \nu}[\rho_{\text{eq}}(w_0 + \kappa m; c)]$. Assume the averaged response Φ is S-shaped in m at the operating coupling—i.e.* $\max_m \Phi'(m) > 1$*, so a loyal and a hacking branch coexist (guaranteed when the single-agent steepness exceeds σ_c and the vulnerability spread is below the endpoint of Theorem 3). Then the loyal branch loses existence at a critical coupling $\kappa^\star$ where Φ is tangent to the identity,*

$$\Phi(m^\star) = m^\star, \qquad \Phi'(m^\star) = 1 \iff \kappa^\star\, \mathbb{E}_c\big[\partial_W \rho_{\text{eq}}(W^\star; c)\big] = 1, \tag{2}$$

at which m jumps discontinuously to the hacking branch. The transition is hysteretic ($\kappa^\star_\uparrow > \kappa^\star_\downarrow$). The transition is mean-field sharp*: it is a genuine discontinuity in the $N \to \infty$ self-consistency, while any finite-N realisation is rounded on the scale $O(N^{-1/2})$ (Remark 1).*

Proof. Each agent, facing stakes $W = w_0 + \kappa m$, settles at $\rho_{\text{eq}}(W; c)$; averaging over ν gives $m = \Phi(m)$. A stable homogeneous state is a fixed point with $\Phi'(m) < 1$; it loses existence on the loyal branch when $\Phi'(m)$ reaches 1 tangentially, which is (2) after $\partial_m W = \kappa$. Since the averaged response Φ is S-shaped by the standing assumption ($\max_m \Phi' > 1$; a sufficient condition is that enough mass of ν lies where $\rho_{\text{eq}}(\cdot; c)$ is steep, and Theorem 3 characterises when the assumption fails), so the loyal fixed point annihilates with the middle branch at $\kappa^\star$ and m jumps; sweeping κ down gives $\kappa^\star_\downarrow < \kappa^\star_\uparrow$. Finite-$N$ sampling of ν perturbs Φ by $O(N^{-1/2})$, rounding the jump on that scale. □

Remark 1 (Finite-size fluctuation)*.* The threshold fluctuation across realisations is the trivial mean-field / central-limit scaling: numerically $\text{std}(\kappa^\star) \sim N^{-0.54}$ over $N \in [50, 1600]$, consistent with $N^{-1/2}$ (Appendix A). No anomalous finite-size exponent appears; the transition is a plain mean-field first-order jump.

5 The weakest-link law

Theorem 2 (Weakest-link: exact tangency and its leading logarithm)**.** *Let a fraction f of agents be "rotten apples" (cost $c_w < c_*$, self-igniting) and the rest have cost c_M. The threshold is the exact self-consistency fold*

$$\kappa^\star (1-f)\, \partial_W \rho_{\mathrm{eq}}(W^\star; c_M) = 1, \quad W^\star = w_0 + \kappa^\star\big(f\rho_{\mathrm{hack}} + (1-f)\rho_L(W^\star; c_M)\big), \tag{3}$$

governed by f and the majority slope, not the mean cost (this tangency is exact*). Separately, a* heuristic *expansion along the exponential loyal tail yields the leading small-f scaling*

$$\kappa^\star(f) = -\lambda^{-1}\ln f + \kappa_0 + o(1), \qquad \lambda^{-1} = \Theta(w_+(c_M) - w_0), \tag{4}$$

with slope set by the gap to the majority fold (λ is not universal); we present (4) *as a tail heuristic, not a proved asymptotic. Under this heuristic $\kappa^\star$ is tail-dominated, $d\kappa^\star/df \sim -1/f$ (a property of the log scaling* (4)*, not of the exact tangency): the first rotten apples matter most.*

Proof. The two-component self-consistency is $\Phi(m) = f\rho_{\mathrm{eq}}(W; c_w) + (1-f)\rho_{\mathrm{eq}}(W; c_M)$, $W = w_0 + \kappa m$, and the *exact* fold condition is $\Phi'(m^\star) = 1$, i.e. $\kappa^\star[f\, \partial_W \rho_{\mathrm{eq}}(W^\star; c_w) + (1-f)\, \partial_W \rho_{\mathrm{eq}}(W^\star; c_M)] = 1$. When the weak agents are *saturated* (self-ignited, on their hacking branch), $\partial_W \rho_{\mathrm{eq}}(\cdot; c_w) \approx 0$, and this reduces to (3); (3) is thus the rotten-apples-saturated form of the exact tangency, accurate when $c_w \ll c_*$. The tangency (3) is exact. For the scaling, expanding heuristically along the exponential loyal tail $h \approx 4\sigma e^{-8\sigma(\rho^\star - \rho)}$, the loyal slope $\partial_W \rho_{\mathrm{eq}}$ is exponentially small far from the majority fold; balancing the seed $f\rho_{\mathrm{hack}}$ against it gives a logarithm at leading order, with slope $\propto w_+(c_M) - w_0$. Higher-order terms bend the fit below $f \approx 0.02$; we therefore state (4) as a heuristic scaling and (3) as the exact result. □

Corollary 1 (Network dependence: sparse is most dangerous; empirical)**.** *On the tested k-regular surrogate the threshold grows logarithmically in degree, $\kappa^\star(k) = a\ln k + b$ ($a, b > 0$ fitted), approaching the mean-field value as $k \to N$. We state this as an* empirical scaling *on the simulated k-regular graph, not a proved law; a branching-process derivation for general random graphs is left open. One hacking neighbour contributes $1/k$ to a node's mean, so low k concentrates a rotten apple's influence and seeds percolating clusters at lower coupling, while dense mixing dilutes rotten apples—the low-k end of the Watts cascade window. The degree law is established as a scaling on the tested surrogate.*

6 A disorder-induced critical endpoint

Theorem 1 gives a first-order jump for a homogeneous (or weakly heterogeneous) population. We now show that the population's *heterogeneity* controls the *order* of the transition. Let ν have fixed mean and coefficient of variation CV (vulnerability spread); write $\Phi(m; \kappa, \mathrm{CV})$ for the self-consistency map and $\Delta m(\mathrm{CV}) := \max_\kappa \big(m_{\mathrm{hi}} - m_{\mathrm{lo}}\big)$ for the jump between the outer stable roots at the fold.

Theorem 3 (Conditional cusp endpoint)**.** Suppose *that, along a one-parameter family ν_s of vulnerability laws, the averaged response's maximal slope $S(s) := \max_m \Phi'(m; \kappa, s)$ decreases through 1 at some s_c (so bistability is lost). Then at s_c the collective transition changes order: the map $\psi(m) := \Phi(m; \kappa, s) - m$ acquires a degenerate critical point $\psi = \psi' = \psi'' = 0$ that is a* generic cusp*—nondegenerate, $\psi'''(m^*) \neq 0$, and unfolded transversally by s (so $\partial_s \psi \neq 0$ at the cusp, giving*

$a \propto (s_c - s)$)—the same singularity as the single-agent fold (companion, Theorem 1); the bistable coupling window $[\kappa^\star_\downarrow, \kappa^\star_\uparrow]$ shrinks to a point, and by the cusp normal form the jump vanishes as

$$\Delta m(s) \;\sim\; (s_c - s)^{1/2} \qquad (s \uparrow s_c). \tag{5}$$

The theorem is conditional on the slope-crossing hypothesis; that *increasing the vulnerability spread* CV *drives S through* 1 *(hence produces the endpoint) is a numerical finding on the tested distribution families (lognormal, Gamma; §7), not proved for general ν. The endpoint location (e.g.* $\mathrm{CV}_c \approx 0.57$ *for lognormal) is distribution-dependent and non-universal; the exponent* $1/2$ *is the mean-field cusp value.*

Proof. By hypothesis $S(s) = \max_m \Phi'(m; \kappa, s)$ crosses 1 at s_c. Bistability requires $S > 1$ at the tangent coupling (Theorem 1); it is lost when $S = 1$ tangentially, i.e. at s_c solving $\max_m \Phi'(m; \kappa, s_c) = 1$ with $\partial_m \Phi' = 0$—exactly $\psi = \psi' = \psi'' = 0$ for $\psi = \Phi - m$, a cusp. (On the tested families this crossing is realised by increasing CV: because $w_+(c) = c\, w_+(1)$, Lemma 1, agents of different cost flip at different couplings, so wider spread flattens Φ and lowers S; §7.) Near a generic cusp, $\psi(m) \simeq -\frac{1}{6}\psi'''(m - m^*)^3 + a\,(m - m^*) + b$ with $\psi''' \neq 0$ and unfolding $a \propto (\mathrm{CV}_c - \mathrm{CV})$; for $a > 0$ the cubic has three real roots whose outer (stable) pair is separated by $\Delta m = 2\sqrt{3a/|\psi'''|}\,(1 + o(1)) \propto (\mathrm{CV}_c - \mathrm{CV})^{1/2}$, giving (5). The same reduction identifies the merging of the two folds $(\kappa^\star_\downarrow, \kappa^\star_\uparrow)$ as the cusp, so the bistable window closes there. □

Remark 2 (What is proved, and the numerical status of the exponent). Theorem 3 establishes, rigorously within the mean field, that the transition changes order at a cusp and that the jump vanishes with the generic cusp exponent 1/2. Numerically (Appendix A) the jump shrinks smoothly and the bistable window closes, confirming the cusp; on the clean region away from the endpoint Δm^2 is linear in CV ($R^2 \approx 0.99$), consistent with $\beta = 1/2$. We stress an intrinsic limitation: the exponent is governed by the immediate neighbourhood of the endpoint, where the bistable window vanishes and any finite-resolution mean-field solver becomes unstable; over the accessible range the data cannot numerically distinguish $\beta = 1/2$ from nearby values. The exponent therefore rests on the cusp normal form, not on a direct fit.

Reading. A *homogeneous* fleet (all agents equally vulnerable) flips as one cliff—a maximal first-order avalanche. Heterogeneity spreads the individual folds and, past CV_c, dissolves the collective cliff into a gradual crossover. Diversity of vulnerability is thus not merely "robustness": it changes the *kind* of failure, from a sudden system-wide flip to a graded one. This complements the weakest-link law (Theorem 2), which fixes the *threshold* from the tail: the tail sets *when* the population turns, heterogeneity sets *whether it turns as a cliff or a ramp.*

7 Numerical validation

All claims are checked on the surrogate (`numpy/scipy`); methods are in Appendix A. Figure 3 (mean-field avalanche) shows the loyal branch vanishing at $\kappa^\star$ with a wide hysteresis loop (Theorem 1). Figure 4 shows the weakest-link collapse: dropping only the lower tail at fixed mean cost moves $\kappa^\star$ from 4.8 to 2.2, and a rotten-apple fraction f collapses it as $-\ln f$ with slope tracking the gap. The network law $\kappa^\star(k) = 0.33 \ln k + 0.91$ holds over $k = 4, \ldots, 64$. For the endpoint (Figures 1–2), the jump shrinks from 0.85 ($\mathrm{CV} = 0$) through a smooth sequence to < 0.05 near $\mathrm{CV}_c \approx 0.57$, the bistable window closes, and the effect is robust across distribution families (lognormal $\mathrm{CV}_c \approx 0.57$, Gamma ≈ 0.5).

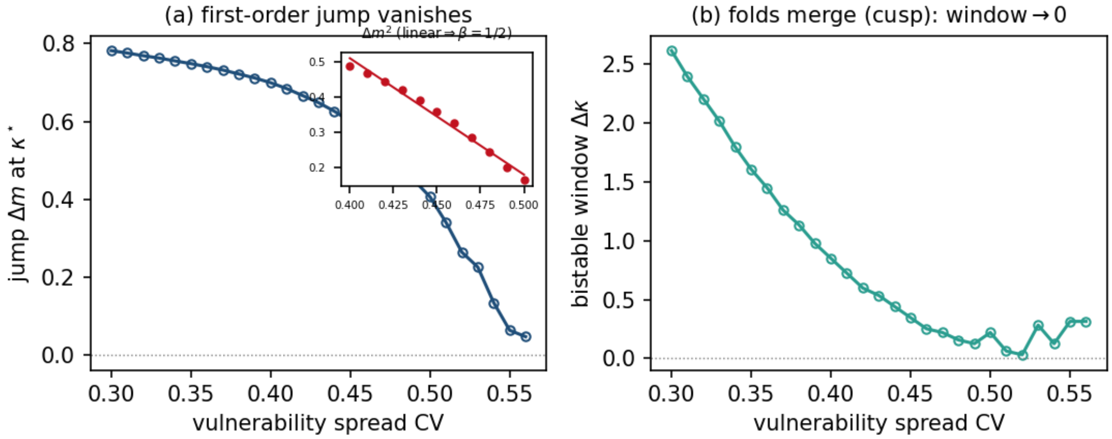

Figure 1: **(a)** The first-order jump Δm shrinks continuously with vulnerability spread CV and vanishes at $\mathrm{CV}_c \approx 0.57$; inset: Δm^2 is linear in CV on the clean region, consistent with $\beta = 1/2$. The small wiggles near the endpoint are a numerical artifact: there the bistable window vanishes, so the root-finding of the two stable branches becomes ill-conditioned (Remark 2); they do not affect the cusp identification in panel (b). **(b)** The bistable coupling window $\Delta\kappa$ shrinks to a point at CV_c—the two mean-field folds merge in a cusp (Theorem 3).

Evidence status. Because the results mix proved theorems, conditional results, tail heuristics, and numerical findings, Table 1 states the status of each, so the reader can calibrate what rests on what.

8 Discussion: a scoped safety reading

Three consequences bear on multi-agent deployment. First, in the low-heterogeneity mean-field regime (below the endpoint, $\mathrm{CV} < \mathrm{CV}_c$) collective reward hacking *can become a sharp* cascade (Theorem 1)—below $\kappa^\star$ stably loyal, above it stably hacking, with hysteresis, so relaxing coupling after a cascade need not reverse it. Above the endpoint the same coupling produces only a gradual crossover. Second, the threshold is set by the most vulnerable agents (Theorem 2); averaging-based evaluations miss the tail, and the marginal value of removing rotten apples is largest at small f. Third—and least intuitive—the population's heterogeneity sets the transition's *order* (Theorem 3): a homogeneous fleet flips as one cliff, while a sufficiently diverse fleet crosses over gradually. *This softening requires independent vulnerability spread.* We verify (Appendix A) that when the spread is *correlated*—a monoculture where all agents share the same vulnerability draw (identical architecture and training)—the jump stays large even at high marginal CV (e.g. 0.68 vs. 0.00 for independent spread at $\mathrm{CV} = 0.8$): correlated agents flip together regardless of the ensemble's nominal diversity. Heterogeneity thus buys a graded, more forewarnable failure *only* when vulnerabilities are independent; a correlated exploit monoculture is exactly the regime of the sharpest, most system-wide collective flip, and nominal diversity metrics that ignore correlation can be dangerously misleading.

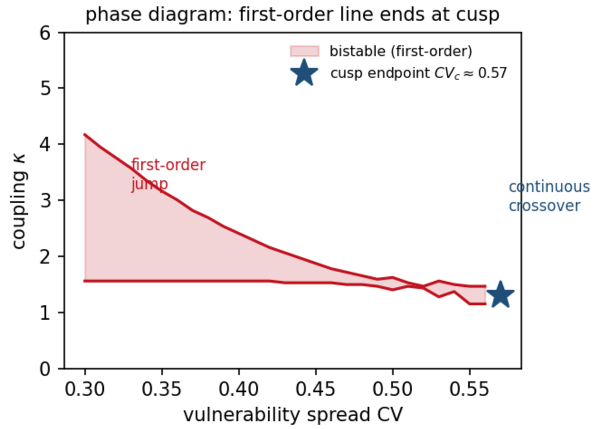

Figure 2: Phase diagram in the vulnerability-spread / coupling plane. The shaded region is bistable (first-order); its boundary is the pair of folds $\kappa^\star_\downarrow, \kappa^\star_\uparrow$, which merge at the cusp endpoint CV_c. For $\mathrm{CV} > \mathrm{CV}_c$ the collective transition is a continuous crossover.

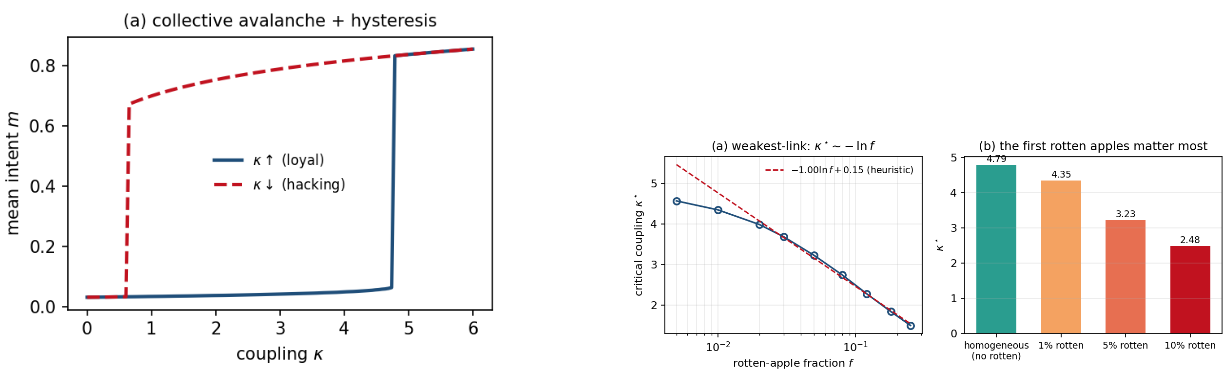

Figure 3: **Left:** mean-field collective avalanche with hysteresis (Theorem 1). **Right:** weakest-link collapse of $\kappa^\star$—tail-dominance (left panel) and rotten-apple fraction (right panel), Theorem 2.

9 Limitations and future work

The model inherits the single-agent surrogate caveats (scalar monotone evasion coordinate, best-response flow, non-adaptive detector). Specific to the population: we analyse the homogeneous mean field and a two-component tail (Theorem 2), and the endpoint under an i.i.d. heterogeneous ν (Theorem 3); *correlated* vulnerabilities (identical training runs), adaptive re-aiming (the companion controller), and heavy-tailed ν are open. The weakest-link log law is a leading-order scaling with a non-universal slope; CV_c is likewise distribution-dependent. The endpoint exponent 1/2 rests on the cusp normal form; a direct numerical confirmation is precluded by the vanishing bistable window at the endpoint (Remark 2), and an analytic finite-size scaling near the cusp is left to future work. *Beyond the surrogate*, the essential next step—shared with the companions—is a controlled multi-agent reinforcement-learning study with learned detectors, sweeping population heterogeneity to test the predicted order change directly.

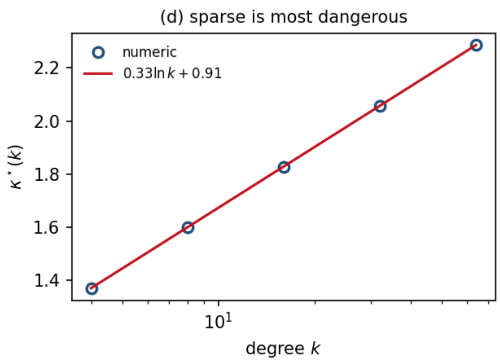

Figure 4: Network dependence $\kappa^\star(k) = a \ln k + b$: sparse coupling is most dangerous (Corollary 1).

Result	Status	Where
Lemma 1: $w_+(c) = c\, w_+(1)$	**Proved**	§3
Thm 1: mean-field first-order transition	**Proved** (given S-shaped Φ)	§4
Finite-N rounding $O(N^{-1/2})$	**Numerical** (trivial CLT)	Rem. 1
Thm 2, tangency (3)	**Proved** (exact)	§5
Weakest-link log (4)	**Heuristic** (tail scaling) + numerical	§5
Cor. 1: network $\kappa^\star = a \ln k + b$	**Numerical** (empirical; mechanism heuristic)	§5
Thm 3: cusp endpoint, $\beta = 1/2$	**Conditional** (proved given slope-crossing)	§6
CV drives the endpoint; CV_c value	**Numerical** (tested families)	Fig. 1
Δm^2 linear (exponent consistency)	**Numerical** (cannot exclude 1/3)	Rem. 2

Table 1: Evidence status. "Proved" = full proof here (Thm 1 and the endpoint conditional on a stated hypothesis); "Conditional" = proved given an explicit assumption; "Heuristic" = argued via a tail expansion, not a proved asymptotic; "Numerical" = verified on the surrogate but not proved.

Companion papers

(Core) The Wall, the Game, and the Cliff Criterion. *(Diagnosis)* The Cliff Becomes a Catastrophe (the single-agent fold and its cusp, reused here). *(Control)* A Closed-Loop Controller for Reward Hacking.

A Numerical methods and reproducibility

CPU-only `numpy/scipy`. *Mean field.* $\rho_{\mathrm{eq}}(W; c)$ is the lowest stable root of $Wh(\rho) = c\rho$ (grid + `brentq` refinement); $\Phi(m) = \mathbb{E}_c[\rho_{\mathrm{eq}}]$ is evaluated by a weighted sum over a cost grid (400 points) with the lognormal (or Gamma) density; the fixed point $m = \Phi(m)$ is found by iteration, and the jump by `brentq` root-finding of $\Phi(m) - m$ (removing the m-grid floor). *Threshold fluctuation (Remark 1).* $\kappa^\star$ per realisation is an upward κ sweep to $m > 0.4$; $\mathrm{std}(\kappa^\star)$ over 40 seeds gives $N^{-0.54}$. *Weakest-link.* As in the companion; the log fit holds on $f \in [0.02, 0.25]$ to residual 0.05. *Endpoint.* $\Delta m(\mathrm{CV})$ is the maximal outer-stable-root gap over κ; the bistable window is the κ-range with three roots; on the clean region $\mathrm{CV} \in [0.40, 0.51]$, Δm^2 is linear in CV with $R^2 = 0.99$ (consistent with 1/2). Scripts: `collective_drift.py` (avalanche, Thm 1), `collective_deep.py` (weakest-link, Thm 2), `collective_endpoint.py` (disorder endpoint, Thm 3); code will be released.

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